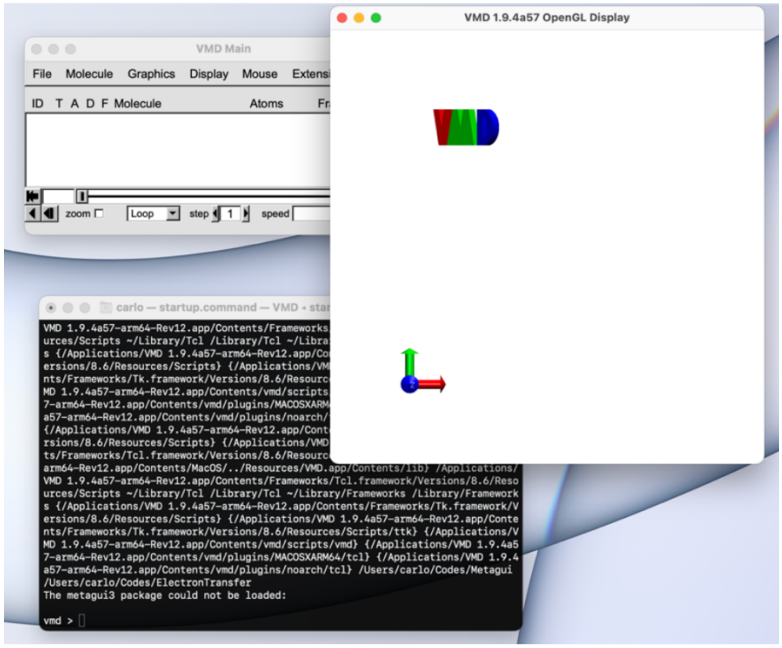
# Using VMD and ChimeraX to visualize the structure of a biomolecule

**AIMS:**
- Familiarize with the visualization options available in VMD
- Familiarize with the visualization options available in ChimeraX

**For the theory, please refer to Lecture 1**
https://github.com/carlocamilloni/Structural-Bioinformatics/blob/main/Notes/01_StructuralBiology.pdf

**VMD**: Visual Molecular Dynamics software is an open-source visualization package developed at the University of Illinois. It is meant to work with both single structure, experimental data, and simulation data. It can also process the data and perform many analyses.
http://www.ks.uiuc.edu/Research/vmd/
Use either the latest version (2.x) or 1.9.3 (particularly on Windows) if the former does not work.

When you start it, you should see something like this: three windows, the bottom one does not matter, the top left one includes all the options available while the right one is the actual visualization panel.

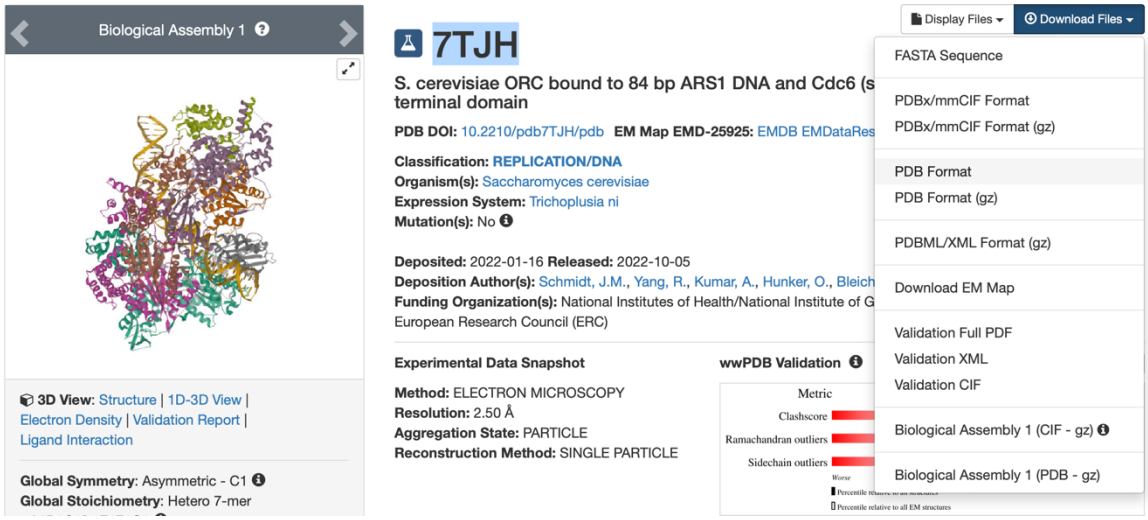


In the following you will visualize and analyze this recent Cryo-EM structure: https://www.rcsb.org/structure/7TJH associated to this work, https://www.nature.com/articles/s41467-022-28695-w "A mechanism of origin licensing control through autoinhibition of S. cerevisiae ORC·DNA·Cdc6." The complex here includes multiple proteins, a DNA molecule, some cofactors and some water molecules. In the following you will learn how to generate some visualisation and analyze some properties.

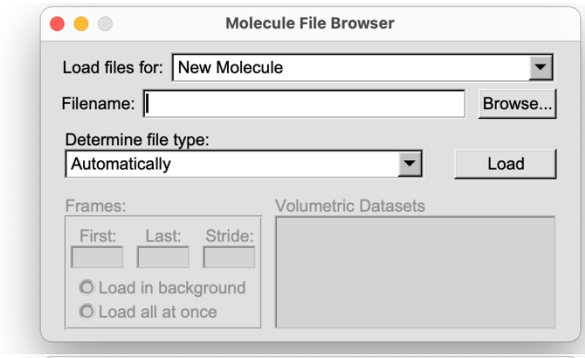
## LOAD A STRUCTURE

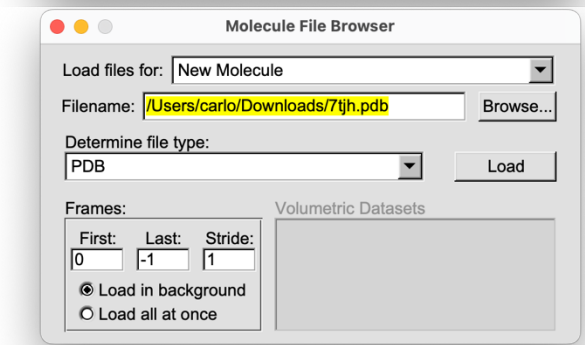
There are two options to open or load a structure in VMD.

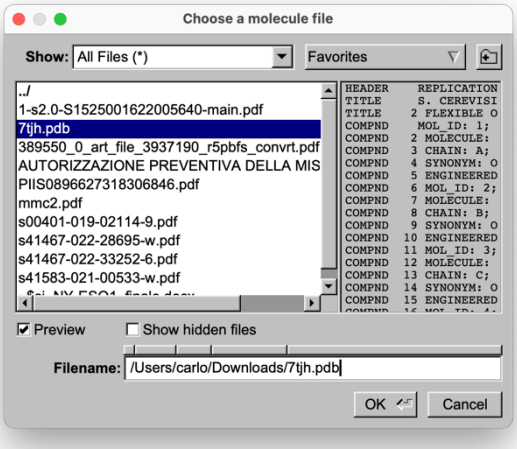
1) Download the PDB structure you want in **PDB format**, in this case this https://www.rcsb.org/structure/7TJH




and then from the menu **File -> New Molecule** will open a new window. From here **Browse…** and try to find your downloaded molecule on your PC, and the **Load** it.

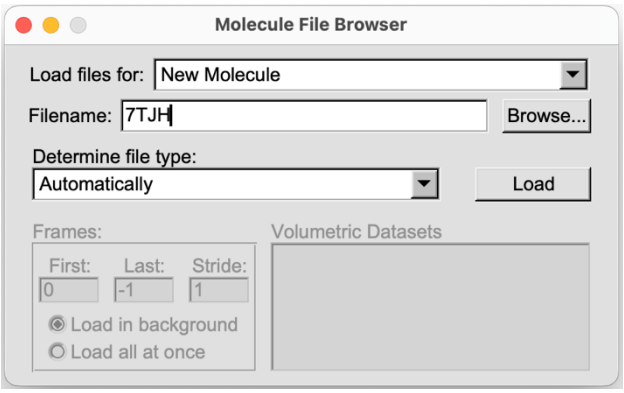


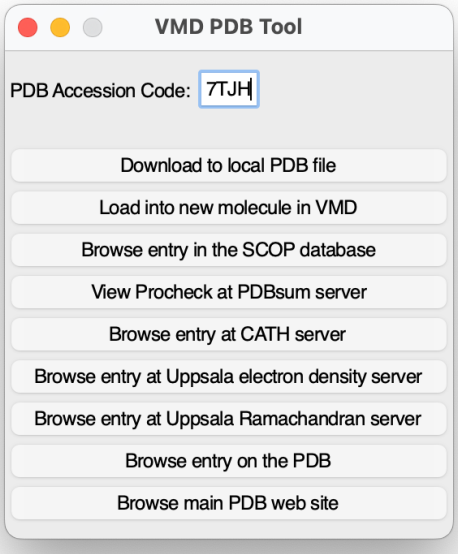



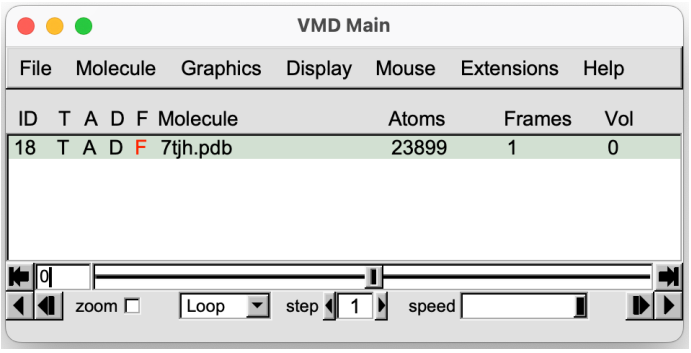
**2)** **ALSO**, you can just write the PDB code in the Filename field and it will actually download the structure.

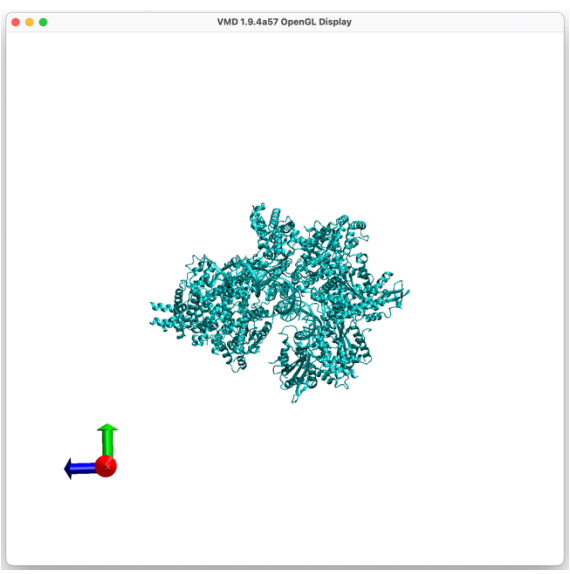
3) Alternatively, from **Extensions -> Data -> PDB Database Query** write the PDB accession code and then **Load into new molecule in VMD**



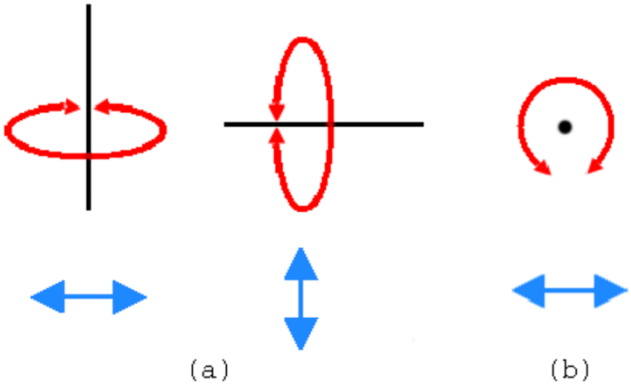




In both cases you should end up with your system loaded and visualize in some default way.




In order to see the 3D structure of our protein, we will use the mouse in multiple modes to change the viewpoint. VMD allows users to rotate, scale and translate the viewpoint of your molecule.




Rotation modes. (a) Rotation axes when holding down the left mouse key. (b) The rotation axis when holding down the right mouse key.

In the OpenGL Display, press the left mouse button down and move the mouse. Explore what happens. This is the rotation mode of the mouse and allows you to rotate the molecule around an axis parallel to the screen.

If you hold down the right mouse button and repeat the previous step, the rotation will be done around an axis perpendicular to your screen. In the **VMD Main window**, look at the **Mouse** menu. Here, you will be able to **switch the mouse mode from Rotation to Translation or Scale modes**.

Choose the **Translation mode** and go back to the OpenGL Display. You can now move the molecule around when you hold the left mouse button down.

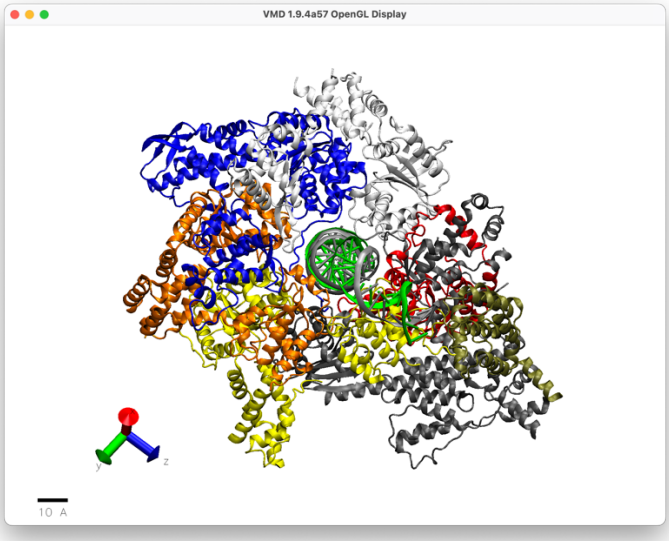
Go back to the the **Mouse menu** and choose the **Scale mode** this time. This will allow you to zoom in or out by moving the mouse horizontally while holding the left mouse button down.

It should be noted that these actions performed with the mouse only change your viewpoint and do not change any of the atomic coordinates associated with the molecule. Each mouse mode is associated to a letter that can be used for fast switching (r, t, and s).

Another useful option is the **Mouse -> Center** menu item. It allows you to specify the point around which rotations are done. Select the **Center** menu item and pick one atom at one of the ends of the protein; The cursor should display a cross. Now, press r, rotate the molecule with the mouse and see how your molecule moves around the point you have selected. In the **VMD Main window**, select the **Display -> Reset View** menu item to return to the default view. **You can also reset the view by pressing the ``=" key when you are in the OpenGL Display window.**

Now let's quickly set some basic view:

1) Set the background color: **Graphics -> Colors -> Display -> Background: choose a color**
2) Set the axes labels color: **Graphics -> Colors -> Axes -> Labels: choose a color**
3) Set the structure view to **Cartoon: Graphics -> Representation**: Choose Drawing Method **NEW CARTOON** and Coloring Method **CHAIN**
4) Set the orthographic view mode: **Display -> Orthographic**
5) Remove depth cueing: **Display -> Depth cueing** (remove the tick if any)
6) Add a ruler bar to get an idea of the size: **Extensions -> Visualisations -> Ruler:** select "scale"




The next pages are taken from the official VMD tutorial

window.

# Graphical Representations

VMD can display your molecule in various ways by the *Graphical Representations* shown in Fig. 7. Each representation is defined by four main parameters: the selection of atoms included in the representation, the drawing style, the coloring method, and the material. The selection determines which part of the molecule is drawn, the drawing method defines which graphical representation is used, the coloring method gives the the color of each part of the representation, and the material determines the effects of lighting, shading, and transparency on the representation. Let's first explore different drawing styles.

## Exploring different drawing styles

**1**

In the VMD Main window, choose the *Graphics* → *Representations...* menu item. A window called Graphical Representations will appear and you will see highlighted in yellow (Fig. 7(a)) the current default representation displaying your molecule.

**2**

In the *Draw Style* tab (Fig. 7(b)) we can change the style (Fig. 7(d)) and color (Fig. 7(c)) of the representation. In this section we will focus in the drawing style (the default is *Lines*).

**3**

Each *Drawing Method* has its own parameters. For instance, change the *Thickness* of the lines by using the controls on the lower right-hand-side corner (Fig. 7(c)) of the Graphical Representations window.

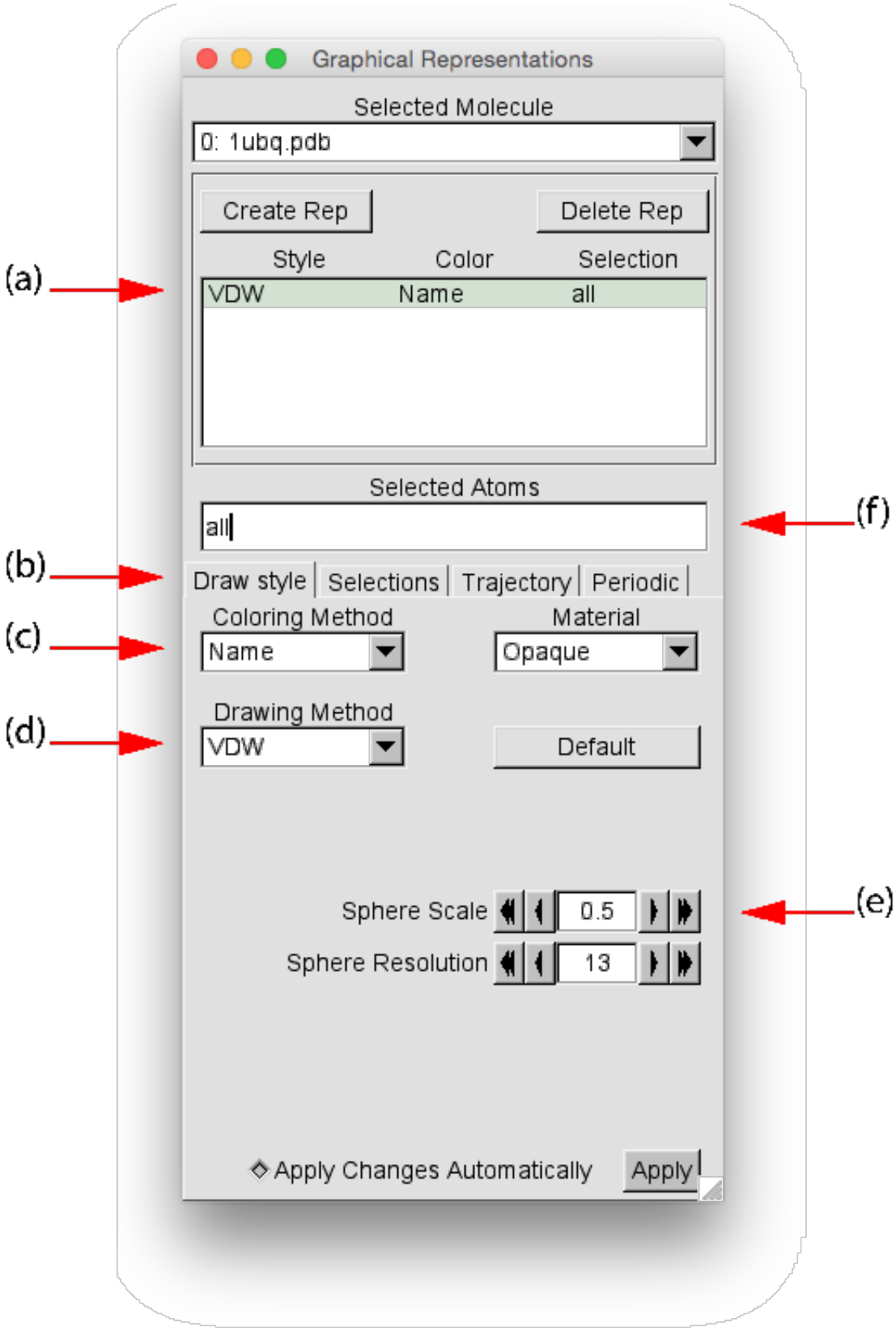



**Figure 7:** The Graphical Representations window.

**4**

Click on the *Drawing Method* (Fig. 7(d)), and you will see a list of options. Choose *VDW* (van der Waals). Each atom is now represented by a sphere, allowing you to see more easily the volumetric distribution of the protein.

**5**

When you choose *VDW* for drawing method, two new controls would show up in the lower right-hand-side corner (Fig. 7(e)). Use these controls to change the *Sphere Scale* to *0.5* and the *Sphere Resolution* to *13*. Be aware that the higher the resolution, the slower the display of your molecule will be.

**6**

Press the *Default* button. This allows you to return to the default properties of the chosen drawing method.

The previous representations allow you to see the micromolecular details of your protein by displaying every single atom. More general structural properties can be demonstrated better by using more abstract drawing methods.

**7**

Choose the *Tube* style under *Drawing Method* and observe the backbone of your protein. Set the *Radius* at *0.8*. You should get something similar to Fig. 8.

**8**

By looking at your protein in the tube drawing method, see if you can distinguish the helices, $\beta$ -sheets and coils present in the protein.

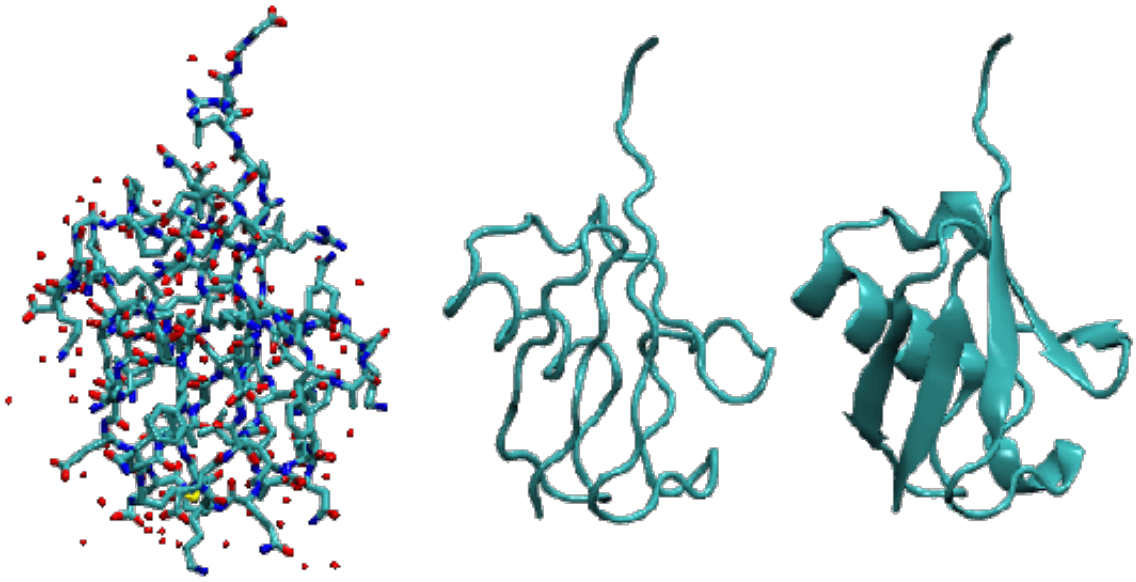
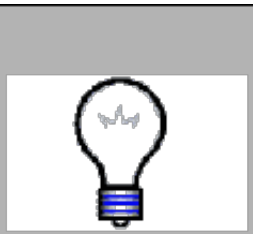


**Figure 8:** Licorice (left), Tube (center) and NewCartoon (right) representations of Ubiquitin



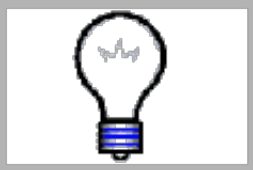
**More representations.** Other popular representations are CPK and Licorice. In CPK, like in old chemistry ball & stick kits, each atom is represented by a sphere and each bond is represented by a thin cylinder (radius and resolution of both the sphere and the cylinder can be modified independently). The Licorice drawing method also represents each atom as a sphere and each bond as a cylinder, but the sphere radius cannot be modified independently.

The last drawing method we will explore is *NewCartoon*. It gives a simplified representation of a protein

based in its secondary structure. Helices are drawn as coiled ribbons, $\beta$ -sheets as solid arrows and all other structures as a tube. This is probably the most popular drawing method to view the overall architecture of a protein.

**9**

In the Graphical Representations window, choose *Drawing Method* $\rightarrow$ *NewCartoon*. You can now easily identify how many helices, $\beta$ -sheets and coils are present in the protein.



**Structure of ubiquitin.** Ubiquitin has three and one half turns of $\alpha$-helix (residues 23 to 34, three of them hydrophobic), one short piece of $3_{10}$-helix (residues 56 to 59) and a mixed $\beta$-sheet with five strands (residues 1 to 7, 10 to 17, 40 to 45, 48 to 50, and 64 to 72) and seven reverse turns. VMD uses the program STRIDE (Frishman et al., *Proteins*, **23**:566, 1995) to compute the secondary structure according to an heuristic algorithm.

## Exploring different coloring methods

Now, let's explore different coloring methods for our representations.

**10**

In the Graphical Representations window, you can see that the default coloring method is *Coloring Method* $\rightarrow$ *Name*. In this coloring method, if you choose a drawing method that shows individual atoms, you can see that they have different colors, i.e: O is red, N is blue, C is cyan and S is yellow.

**11**

Choose *Coloring Method* $\rightarrow$ *ResType* (Fig. 7(c)). This allows you to distinguish non-polar residues (white), basic residues (blue), acidic residues (red) and polar residues (green).

**12**

Select *Coloring Method* $\rightarrow$ *Secondary Structure* (Fig. 7(c)) and confirm that the *NewCartoon* representation displays colors consistent with secondary structure.

## Displaying different selections

You can also only display parts of the molecule that you are interested in by specifying you selection in the Graphical Representations window (Fig. 7(f)).

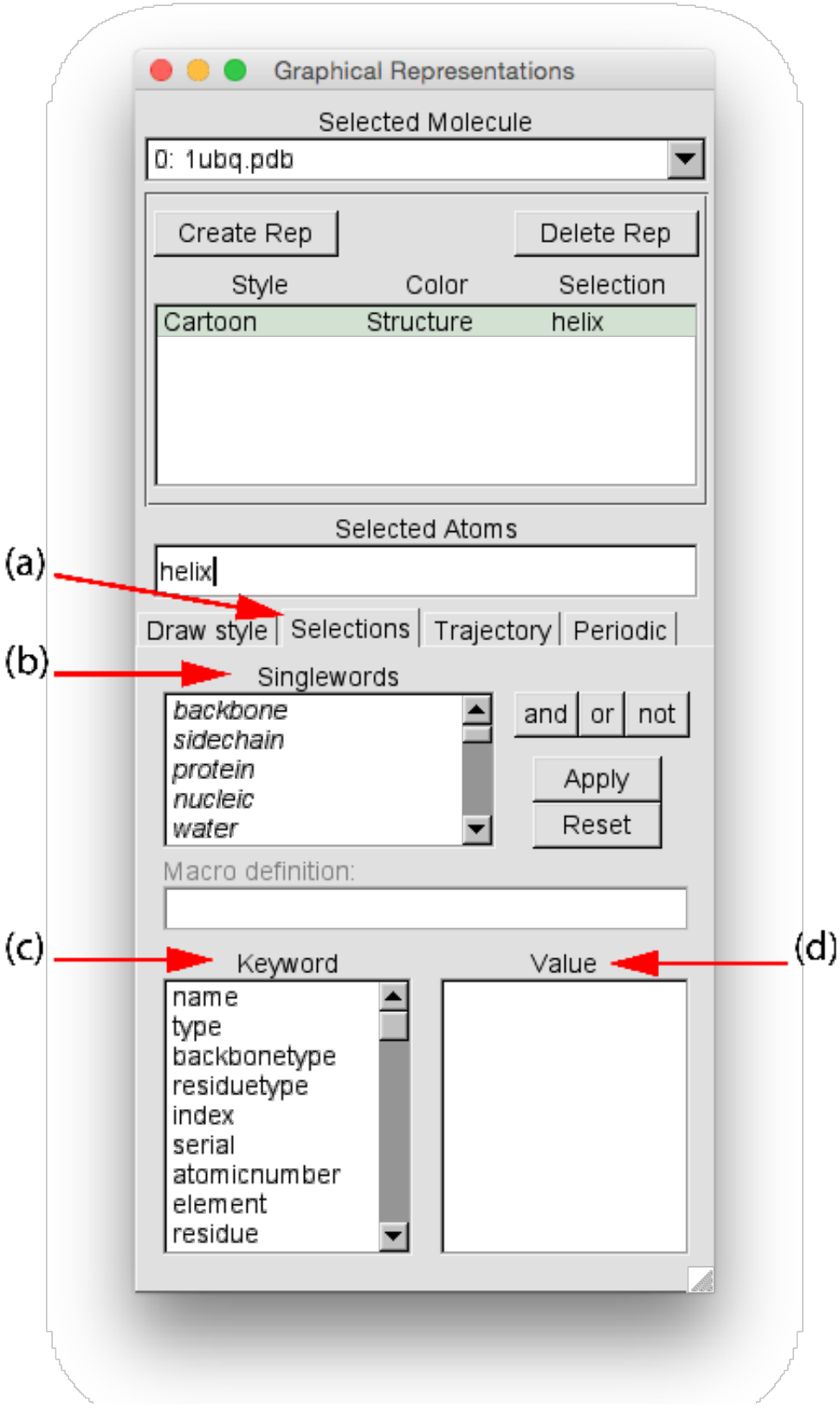



**Figure 9:** Graphical Representations window and the *Selections* tab.

**13**

In the Graphical Representations window, there is a *Selected Atoms* text entry (Fig. 7(f)). Delete the word `all`, type `helix` and press the *Apply* button or hit the Enter/Return key on your keyboard (remember to do this whenever you change a selection). VMD will show just the helices present in our molecule.

**14**

In the Graphical Representations window choose the *Selections* tab (Fig. 9(a)). In section *Singlewords* (Fig. 9(b)), you will find a list of possible selections you can type. For instance, try to display $\beta$ -sheets instead of helices by typing the appropriate word in the *Selected Atoms* text entry.

Combinations of boolean operators can also be used when writing a selection.

**15**

In order to see the molecule without helices and $\beta$ -sheets, type the following in *Selected Atoms*:

`(not helix) and (not betasheet)`. Remember to press the *Apply* button or hit the Enter/Return key on your keyboard.

**16**

In the section *Keyword* (Fig. 9(c)) of the *Selections* tab, you can see properties that can be used to select parts of a protein with their possible values. Look at possible values of the keyword *resname* (Fig. 9(d)). Display all the lysines and glycines present in the protein by typing `(resname LYS) or (resname GLY)` in the *Selected Atoms*. Lysines play a fundamental role in the configuration of polyubiquitin chains.

**17**

Now, change the current representation's *Drawing Method* to *CPK* and the *Coloring Method* to *ResName* in the *Draw Style* tab. In the screen you will be able to see the different Lysines and Glycines.

**18**

In the *Selected Atoms* text entry type `water`. Choose *Coloring Method* $\rightarrow$ *Name*. You should see the 58 water molecules (in fact only the oxygens) present in our system.

**19**

In order to see which water molecules are closer to the protein you can use the command `within`. Type `water and within 3 of protein` for *Selected Atoms*. This selects all the water molecules that are within a distance of 3 angstroms of the protein.

**20**

Finally, try typing the following selections in *Selected Atoms*:

**Table 1:** Example atom selections.

| Selection | Action |
|---|---|
| `protein` | Shows the Protein |
| `resid 1` | The first residue |
| `(resid 1 76) and (not water)` | The first and last residues |
| `(resid 23 to 34) and (protein)` | The $\alpha$ -helix |

## Creating multiple representations

The button *Create Rep* (Fig. 10(a)) in the Graphical Representations window allows you to create multiple representations. Therefore, you can have a mixture of different selections with different styles and colors, all displayed at the same time.

**21**

For the current representation, in *Selected Atoms* type `protein`, set the *Drawing Method* to *NewCartoon* and the *Coloring Method* to *Secondary Structure*.

**22**

Press the *Create Rep* button (Fig. 10(a)). You should see that a new representation is created. Modify the new representation to get *VDW* as the *Drawing Method*, *ResType* as the *Coloring Method*, and `resname LYS` as the current selection.

**23**

Repeating the previous procedure, create the following two new representations:

**Table 2:** Example representations.

| Selection | Coloring Method | Drawing Method |
|---|---|---|
| `water` | *Name* | *CPK* |
| `resid 1 76 and name CA` | *ColorID* $\rightarrow$ *1* | *VDW* |

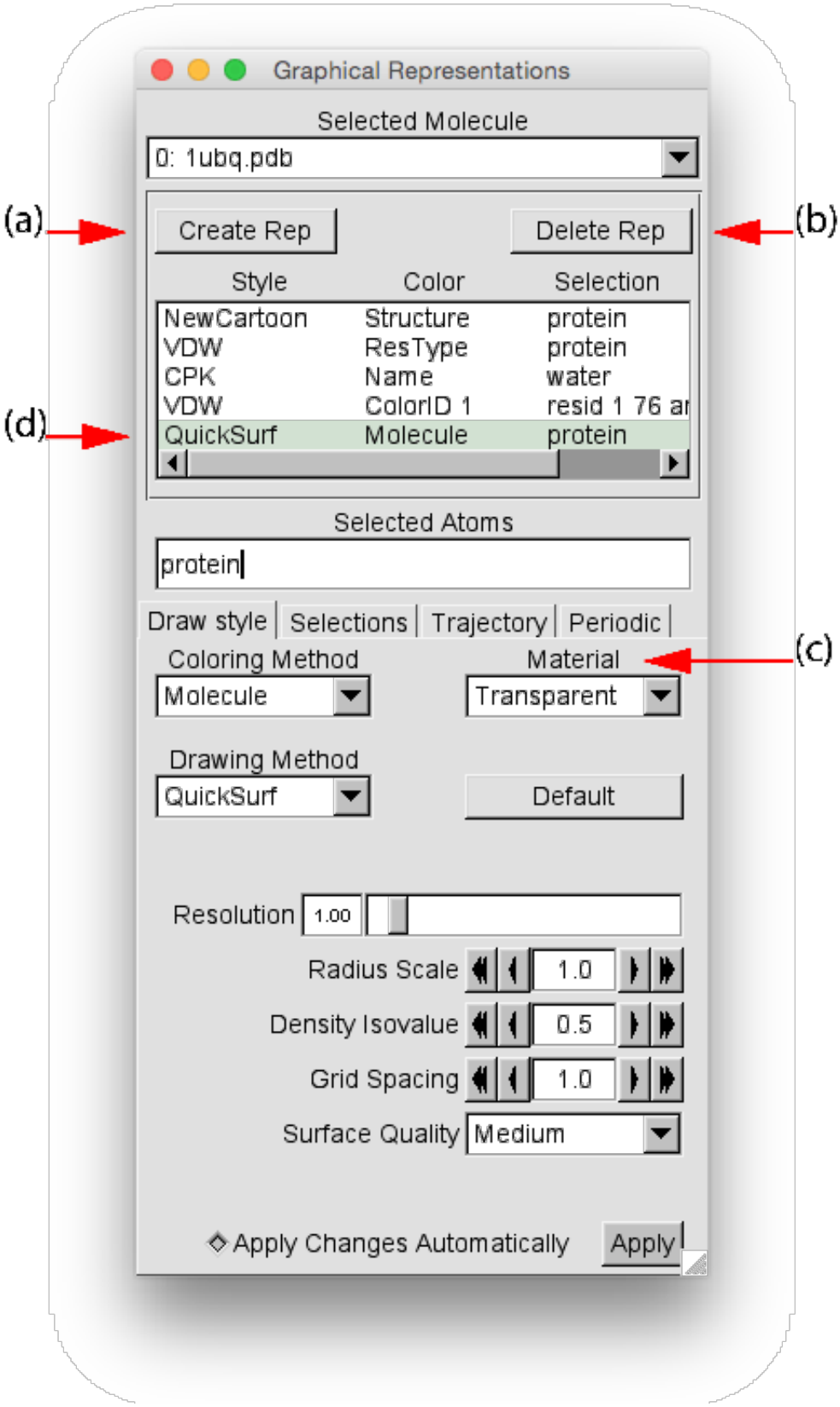



**Figure 10:** Multiple Representations of Ubiquitin.

**24**

Create the last representation by pressing again the *Create Rep* button. Select *Drawing Method* → *QuickSurf* for drawing method, *Coloring Method* → *Molecule* for coloring method, and type `protein` in the *Selected Atoms* entry. Set the QuickSurf radius scale parameter to 0.7. For this last representation choose *Transparent* in the *Material* pull-down menu (Fig. 10(c)). This representation

shows protein's volumetric surface in transparent.

**25**

Note that you can select and modify different representations you have created by clicking on a representation to highlight it in yellow. Also, you can switch each representation on/off by double-clicking on it. You can also delete a representation by highlighting it and clicking on the *Delete Rep* button (Fig. 10(b)). At the end of this section, your Graphical Representations window should look similar to Fig. 10.

# Sequence Viewer Extension

When dealing with a protein for the first time, it is very useful to find and display different amino acids quickly. The sequence viewer extension allows you to view the protein sequence, as well as picking and displaying one or more residues of your choice easily.

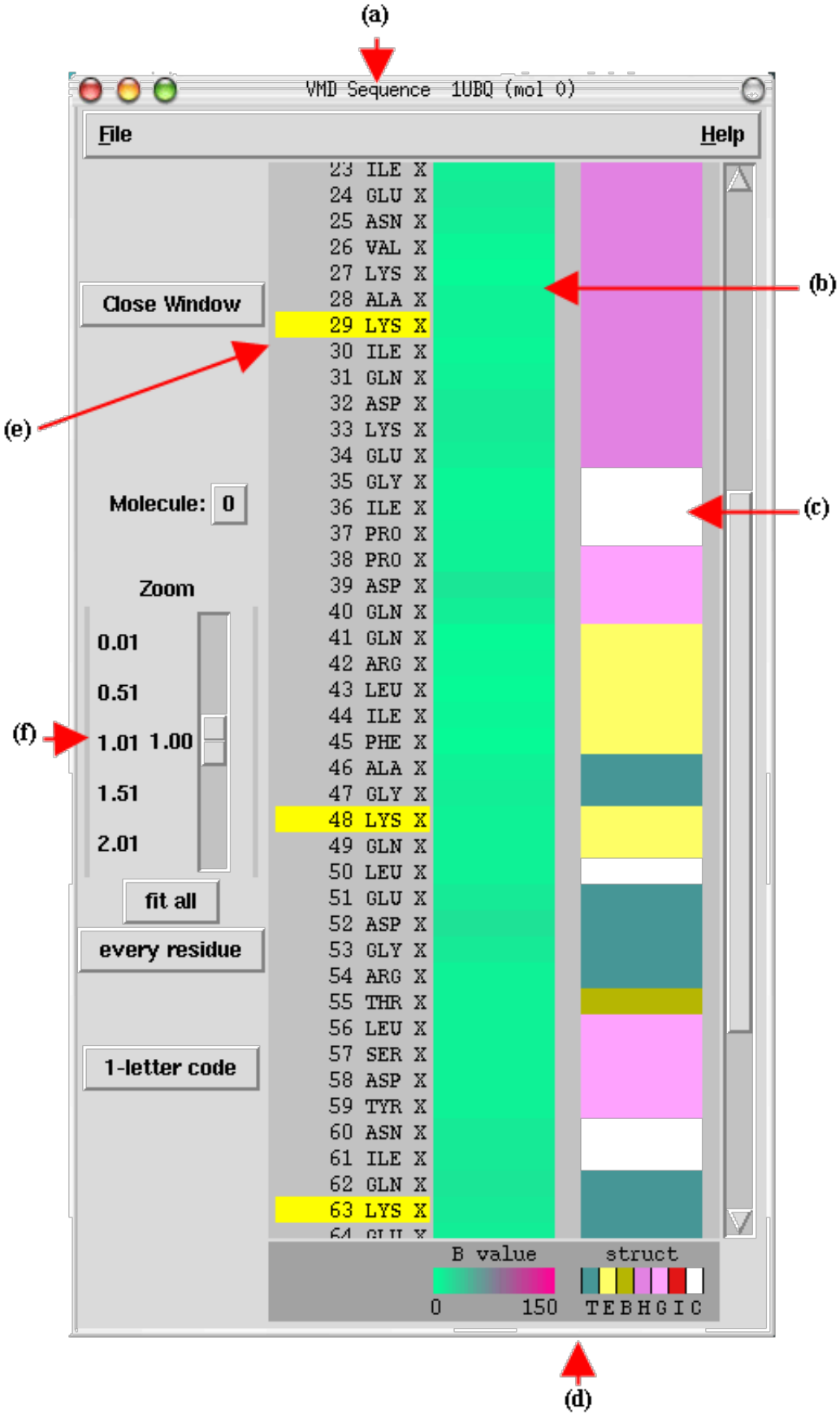



**Figure 11:** VMD Sequence window.

**1**

In the VMD Main window, choose the *Extensions* → *Analysis* → *Sequence Viewer* menu item. A

window (Fig. 11(a)) with a list of the amino acids (Fig. 11(e)) and their properties (Fig. 11(b)&(c)) will appear in your screen.

**2**

With the mouse, try clicking on different residues in the list (Fig. 11(e)) and see how they are highlighted. In addition, the highlighted residue will appear in your OpenGL Display window in yellow and bond drawing method, so you can visualize its location within the protein easily.

**3**

Using the *Zoom* controls (Fig. 11(f)) you can display the entire list of residues in the window. This is especially useful for larger proteins

**4**

To pick multiple residues, hold the shift key and click on the mouse button. Try highlighting residues 11, 48, 63 and 29 (Fig. 11(e)).

**5**

Look at the Graphical Representations window, you should find a new representation with the residues you have selected using Sequence Viewer Extension. You can modify, hide or delete this representation similar to what you have done before.

Information about residues is color-coded (Fig. 11(d)) in columns and obtained from STRIDE. The *B-value* column (Fig. 11(b)) shows the B-value field (temperature factor). The *struct* column shows secondary structure (Fig. 11(d)), where each letter means:

**Table 3:** Secondary Structure codes used by STRIDE.

| | |
|---|---|
| T | Turn |
| E | Extended conformation ( $\beta$ -sheets) |
| B | Isolated bridge |
| H | Alpha helix |
| G | 3-10 helix |
| I | Pi helix |
| C | Coil |

# Saving Your Work

The viewpoints and representations that you have created using VMD can be saved as a VMD state. This VMD state contains all the information needed to reproduce the same VMD session without losing what you have done.

**1**

Go to the OpenGL Display window, use your mouse to find a nice view of the protein. We will save this viewpoint using VMD ViewMaster.

**2**

In the VMD Main window, select *Extension* → *Visualization* → *ViewMaster*. This will open the VMD ViewMaster window.

**3**

In the VMD ViewMaster window, click on the *Create New* button. Now you have saved your OpenGL Display view point.

**4**

Go back to your OpenGL Display window, use your mouse to find another nice view. If you want you can also add/delete/modify a representation in the Graphical Representations window. When you have found a good view, you can again save it by returning to the VMD ViewMaster window and clicking on the *Creat New* button.

**5**

Create as many views as you like by repeating the previous step. You can see that in the VMD ViewMaster window, all of your viewpoints are displayed as thumbnails. You can go to a previously-saved viewpoint by clicking on its thumbnail.

**6**

Let's now save the entire VMD session. In the VMD Main window, choose the *File* → *Save Visualization State* menu item. Write an appropriate name (e.g., `myfirststate.vmd`) and save it. The VMD state file `myfirststate.vmd` contains all the information you need to restore your VMD session, including the viewpoints and the representations.

To load a saved VMD state, start a new VMD session and in the VMD Main window choose *File* → *Load State*.

**7**

Quit VMD.

# The Basics of VMD Figure Rendering

One of VMD's many strengths is its ability to render high-resolution, publication-quality molecule images. In this section we will introduce some basic concepts of figure rendering in VMD.

## Setting the display background

Before you render a figure, you want to make sure you set up the OpenGL Display background the way you want. Nearly all aspects of the OpenGL Display are user-adjustable, including background color.

**1**

Start a new VMD session. Load the `1ubq.pdb` file in the `vmd-tutorial-files` directory by following the steps in Section 1.1.

**2**

In the VMD Main window, choose *Graphics* → *Colors*.... The Color Controls window should show

up. Look through the *Categories* list. All display colors, for example, the colors of different atoms when colored by name, are set here.

**3**

Now we will change the background color. In *Categories*, select *Display*. In *Names*, select *Background*. Finally, choose *8 white* in *Colors*. Your OpenGL Display now should have a white background.

**4**

When making a figure, we often don't want to include the axes. To turn off the axes, select *Display* → *Axes* → *Off* in the VMD Main window.

## Increasing resolution

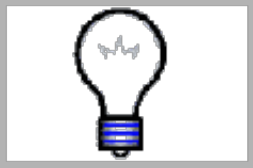
Most VMD graphical representations are drawn with an adjustable resolution, allowing users to balance geometric detail with interactive drawing speed.

**5**

Open the Graphical Representation window via *Graphics* → *Representations*... in the VMD Main menu. Modify the default representation to show just the protein, and display it using the VDW drawing method.

**6**

Zoom in on one or two of the atoms, either by using the scroll wheel on your mouse, or by using *Mouse* → *Scale Mode* (shortcut `s`).



**OpenGL clipping.** You might notice that as you zoom or move the view close up into an atom, it might be cut open or chopped away by an invisible clipping plane that lies just ahead or coincident with the position of the camera. The so-called "near" clipping plane in OpenGL is a detail of its implementation, and it can be made nearly (but not quite) zero to minimize its impact. You can move the molecule toward or away from you (the camera position) so that the clipping doesn't impact the molecular scene in a detrimental way by doing the following: switch your mouse mode to the Translate mode, either by pressing the shortcut key "t" in the OpenGL window or by selecting Mouse → Translate Mode, and drag the mouse in the OpenGL window while holding down the right mouse key. You can now move the molecule toward or away from you (the camera). If moving the molecular scene toward or away from the camera is ineffective, the default OpenGL "near" clipping plane can be adjusted as follows: in the VMD Main window, choose Display → Display Settings...; in the Display Settings window that shows up, you can see many OpenGL display options you can adjust; decrease the value for Near Clip, this will move the OpenGL clipping plane closer, allowing you to zoom or translate the camera into individual atoms without clipping them off.

**7**

Notice that with the default resolution setting, the ``spherical'' atoms aren't looking very spherical. In the Graphical Representations window, click on the representation you set up before for the protein to highlight it in yellow. Try adjusting the *Sphere Resolution* setting to something higher, and see what a difference it can make. (See Fig. 12.)

Many of the drawing methods have a resolution setting. Try a few different drawing methods and see how you can easily increase their resolutions. When producing images, you can raise the resolution until it stops making a visible difference.

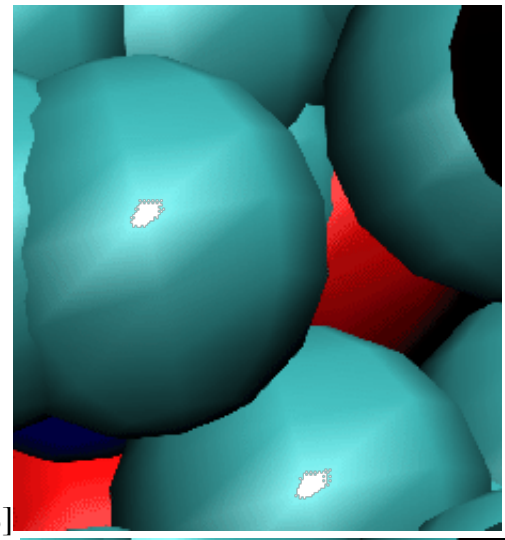
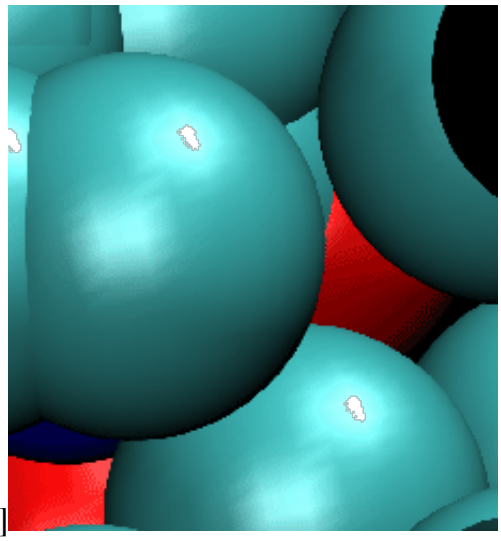
[Low resolution: *Sphere Resolution* set to 8]  [High resolution: *Sphere Resolution* set to 28] 

**Figure 12:** The effect of the resolution setting.

## Colors and materials

**8**

You may have noticed the *Material* menu in the Graphical Representations window (which by default is drawn with *Opaque* material). Choose the protein representation you made before, and experiment with the different materials in the *Material* menu.

**9**

Besides the pre-defined materials in the *Material* menu, VMD also allows users to create their own materials. To make a new material, in the VMD Main window choose *Graphics* → *Materials*.... In

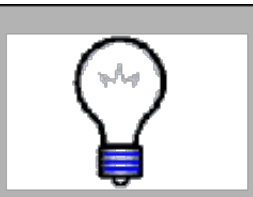
the Materials window that appears, you'll see a list of the materials you just tried out, and their adjustable settings. Click the *Create New* button. A new material, e.g. *Material23*, will be created. Give it the following settings:

**Table 4:** Example user-defined material.

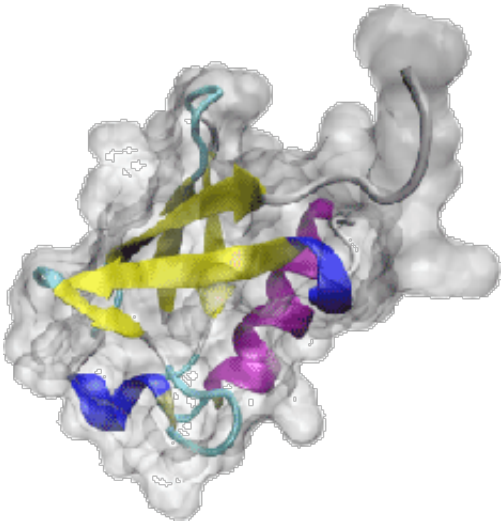
| Setting | Value |
|---|---|
| *Ambient* | 0.30 |
| *Diffuse* | 0.30 |
| *Specular* | 0.90 |
| *Shininess* | 0.50 |
| *Opacity* | 0.95 |

**10**

Go back to the Graphical Representations window. In the *Material* menu, you can see that *Material23* is now on the list. Try using *Material23* for a representation and see what it looks like.



**GLSL.** Now is a good time to try out the GLSL Render Mode, if your computer supports it. In the VMD Main window, choose Display → Rendermode → GLSL. This mode uses your 3D graphics card to render the scene with real-time raytracing of spheres and alpha-blended transparency; see Fig. 13 for an example.



[The default transparent material.] [A user-defined material.]

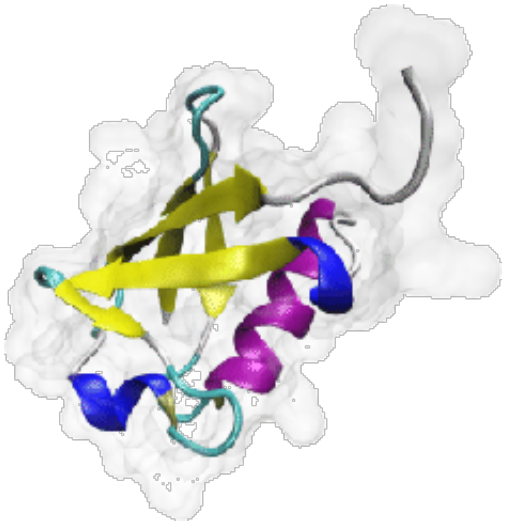


**Figure 13:** Examples of different material settings.

**11**

If your computer supports GLSL Render Mode, you can try to reproduce Fig. 13(b). First turn on the GLSL rendering mode by selecting *Display* → *Rendermode* → *GLSL* in the VMD Main window.

**12**

Modify *Material23* to be more transparent by entering the following values in the Materials window:

**Table 5:** Example of a more transparent material.

| Setting | Value |
|---|---|
| *Ambient* | 0.30 |
| *Diffuse* | 0.50 |
| *Specular* | 0.87 |
| *Shininess* | 0.85 |
| *Opacity* | 0.11 |

**13**

Hide all of your current representations and create the following two representations:

**Table 6:** Example of representations drawn with different materials.

| Selection | Coloring Method | Drawing Method | Material |
|---|---|---|---|
| `protein` | *Structure* | *NewCartoon* | *Opaque* |
| `protein` | *ColorID* → *8 white* | *QuickSurf* | *Material23* |

## Depth perception

Since the systems we are dealing with are three-dimensional, VMD has multiple ways of representing the third dimension. In this section, we will explore how to use VMD to enhance or hide depth perception.

**14**

The first thing to consider is the projection mode. In the VMD Main window, click the *Display* menu. Here we can choose either *Perspective* or *Orthographic* in the drop-down menu. Try switching between *Perspective* and *Orthographic* projection modes and see the difference (Fig. 14).

In perspective mode, things nearer the camera appear larger. Although perspective projection provides strong size-based visual depth cues, the displayed image will not preserve scale relationships or parallelism of lines, and objects very close to the camera may appear distorted. Orthographic projection preserves scale and parallelism relationships between objects in the displayed image, but greatly reduces depth perception.

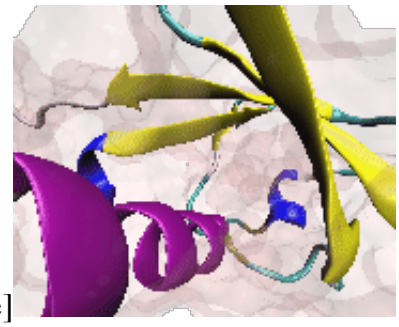
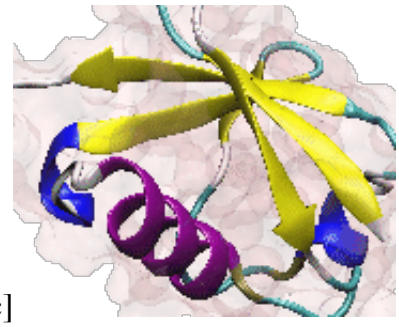
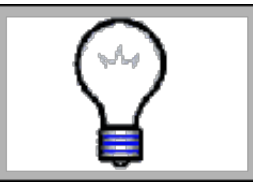
[Perspective]  [Orthographic] 

**Figure 14:** Comparison of perspective and orthographic projection modes.



**Orthographic versus Perspective Mode.** Orthographic mode tends to be more useful for analysis, because alignment is easy to see, while perspective mode is often used for producing figures and stereo images.

Another way VMD can represent depth is through the so-called ``depth cueing''. Depth cueing is used to enhance three-dimensional perception of molecular structures, particularly with orthographic projections.

**15**

Choose *Display* → *Depth Cueing* in the VMD Main window. When depth cueing is enabled, objects further from the camera are blended into the background. Depth cueing settings are found in *Display* → *Display Settings*.... Here you can choose the functional dependence of the shading on distance, as well as some parameters for this function. To see the effect better, you might want to hide the representation with the *QuickSurf* drawing method.

**16**

Finally, VMD can also produce stereo images. In the VMD Main window, look at the *Display* → *Stereo* menu, showing many different choices. Choose *SideBySide* (remember to return to Perspective mode for a better result). You should get something like Fig. 15

**17**

Turn off stereo image by selecting *Display* → *Stereo* → *Off* in the VMD Main window. Also turn off depth cueing by unselecting the *Display* → *Depth Cueing* checkbox in the VMD Main window.

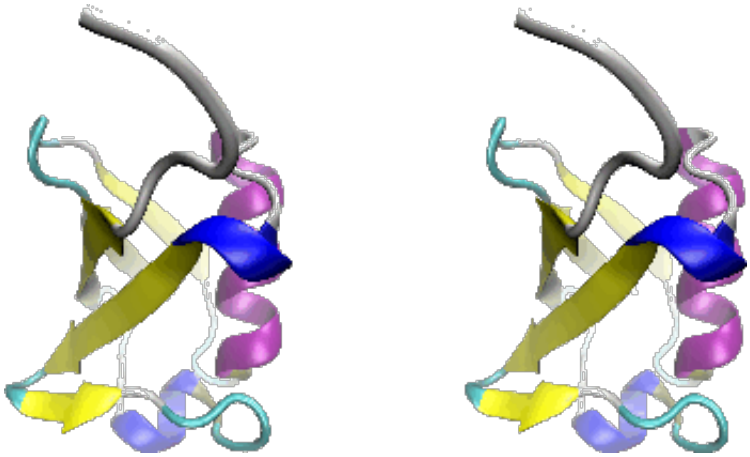


**Figure 15:** Stereo image of the ubiquitin protein. Shown here with *Cue Mode* = *Linear*, *Cue Start* = 1.5, and *Cue End* = 2.75.

## Rendering

By now we've seen some techniques for producing nice views and representations of the molecule loaded in VMD. Now, we'll explore the use of the VMD built-in snapshot feature and external rendering programs to produce high quality images of your molecule. The ``snapshot'' renderer saves the on-screen image in your OpenGL window and is often adequate for use in drafts of presentations, movies, and small figures. When one desires higher quality images for publication, renderers such as Tachyon (and its hardware-accelerated variants) and POV-Ray are much better choices due to their improved rendering quality and support for advanced lighting and shading. One of the key benefits of Tachyon and POV-Ray vs. the ``snapshot'' renderer is that they can directly render curved geometric primitives such as spheres and cylinders, eliminating the need to be concerned with the ``resolution'' parameters for most graphical representations as described in Sec. 1.6.2.

Sophisticated renderers such as Tachyon and POV-Ray perform challenging rendering calculations in exchange for the high quality images they produce. When rendering very high resolution images in combination with large numbers of transparent surfaces, shadows, ambient occlusion lighting, and other advanced features, the rendering time required by Tachyon or POV-Ray can start to become a noteworthy consideration, particularly in the context of movie rendering. VMD includes an I/O optimized Tachyon rendering path called TachyonInternal that completely avoids writing VMD scene files and constituent geometry to disk, instead rendering directly from the in-memory VMD scene and writing out only the final rendered image. The use of TachyonInternal provides a large performance benefit when rendering large geometrically complex VMD scenes and is the basis for even higher performance hardware-optimized Tachyon variants. VMD optionally incorporates two hardware-optimized ``light weight'' versions of the Tachyon rendering engine that use GPU-acceleration and/or CPU vectorization to achieve higher performance than the fully-general cross-platform version of Tachyon. These hardware-optimized

variants of Tachyon are referred to as TachyonLOptiX and TachyonLOptiXInternal (NVIDIA GPU-acceleration), and TachyonLOSPRay and TachyonLOSPRayInternal (Intel CPU vectorization), each of which offer substantial performance gains (frequently 2 × up to as much as 10 × faster) over the full-featured cross-platform Tachyon renderer.

**18**

Hide or delete all your previous representations, and create the new representations shown in Table 7.

**Table 7:** Example representations.

| Selection | Coloring Method | Drawing Style | Material |
|---|---|---|---|
| `protein and not resid 72 to 76` | *Structure* | *NewCartoon* | *Opaque* |
| `protein and helix and name CA` | *ColorID → 8* | *QuickSurf* | *Material 23* |
| | | | (or *Material* |
| | | | with another number) |
| `resname GLY and not resid 72 to 76` | *ColorID → 7* | *VDW* | *Opaque* |
| `resname LYS` | *ColorID → 18* | *Licorice* | *Opaque* |

**19**

Rendering is very simple in VMD. Once you have the scene set the way you like it in the OpenGL window, simply choose *File → Render...* in the VMD Main window. The File Render Controls window will appear on your screen.

**20**

The File Render Controls allows you to choose which renderer you want to use and the file name for your image. For our first try, let's select *snapshot* for the rendering method, type in a filename of your choice, and click *Start Rendering*.

**21**

If you are using a Mac or a Linux machine, an image-processing application might open automatically that shows you the molecule you have just rendered using *snapshot*. If this is not the case, use any image-processing application to take a look at the image file. Close the application when you are done to continue using VMD.

**22**

Try to render again using different rendering method, particularly TachyonInternal and POV3. If you're using a GPU-accelerated hardware platform, try using one of the TachyonLOptiX renderers, and if running on appropriate Intel CPU hardware, try TachyonLOSPRay. Compare the quality of the images created by different renderers.



**Renderers.** The snapshot renderer saves exactly what is already showing in your display window — in fact, if another window overlaps the display window, it may distort the overlapped region of the image. The other renderers (e.g. POV3 and Tachyon) reprocess everything, so it may not look exactly as it does in the OpenGL window. In particular, they don't "clip", or hide, objects very near the camera. If you select Display → Display Settings... in the VMD Main window, you can set Near Clip to 0.01 to get a better idea of what will appear in your rendering.

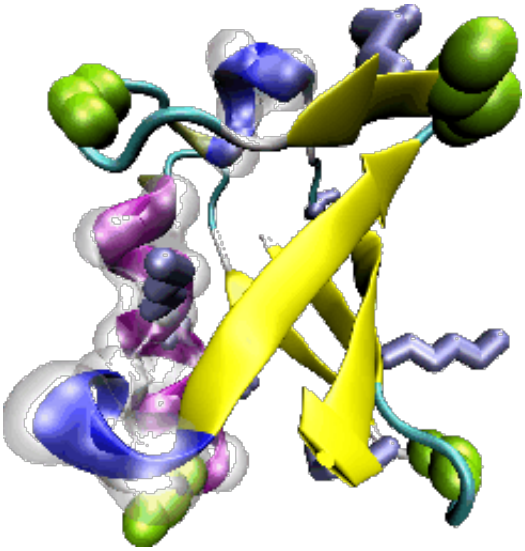


**Figure 16:** Example of a POV-Ray 3.6 (POV3) rendering.

**23**

You have learned the basics of VMD. Quit VMD.

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Next Up Previous

**Next:** Trajectories and Movie Making **Up:** VMD Tutorial **Previous:** Introduction

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# EXERCISES

Now try to generate the following views and save the resulting visualization as images

1) Visualise for one of the ATP/MG2+ molecules its binding pose (**hint:** *same residue as (protein within 5 of resname ATP or ions) or resname ATP or ions* allows you to select all residues that within 5 angstrom of an ATP residue or an ion as well as the ATP and the ion themselves), can you comment on the interactions at play?
2) The region 323 to 408 of ORC5 is responsible for the bending of the DNA, generate an image showing this interaction.

# Using ChimeraX to visualize the structure of a biomolecule

**AIMS:**

- Familiarize with UCSF ChimeraX, its command line and its graphical tools
- Analyze a binding site and map sequence conservation onto a structure

UCSF ChimeraX is the next-generation visualization program from the RBVI at UC San Francisco, the successor of Chimera. It is free for academic use and handles atomic structures, cryo-EM and X-ray density maps, sequence alignments and predicted models. Most actions are typed in the **command line** at the bottom of the window (`help style` opens the documentation of any command); the same actions are also available from menus, toolbar icons and right-click context menus. Download the latest release for your OS from https://www.cgl.ucsf.edu/chimerax/download.html.

## LOAD A STRUCTURE AND SET A BASIC VIEW

Type `open 7tjh`: the structure is fetched directly from the PDB (mmCIF format), no download needed. The **Log** lists the chains, the non-standard residues (ATP, $Mg^{2+}$, ...) and the assemblies; clicking an entry selects it. Mouse: left-drag rotates, middle/right-drag translates, the wheel zooms, **Ctrl**-click selects (**Shift-Ctrl** adds, up/down arrows expand/shrink the selection); `view` resets the view (as "=" in VMD). The steps you did in VMD, and the VMD selection language, translate as follows:

| VMD step | ChimeraX command |
|---|---|
| Background colour | `set bgColor white` |
| NewCartoon, colour by chain | `cartoon; color bychain` |
| Orthographic view | `camera ortho` |
| Depth cueing off | `lighting depthCue false` |
| Nicer look (ChimeraX only) | `lighting soft; graphics silhouettes true` |
| Sequence Viewer | `sequence chain /E` |
| Render an image | `save 7tjh.png width 2000 supersample 3` |
| Save Visualization State | `save 7tjh.cxs` |

| VMD selection | ChimeraX specification |
|---|---|
| `chain E` | `/E` |
| `resid 323 to 408` | `:323-408` |
| `chain E and resid 323 to 408` | `/E:323-408` |
| `resname ATP` | `:ATP` |
| `name CA` | `@CA` |
| `protein, nucleic, water, ions` | `protein, nucleic, solvent, ions` |
| `not, and, or` | `~ & \|` |
| `same residue as (within 5 of resname ATP)` | `:ATP :<5` |

## TUTORIALS

Work through the following official tutorials in order. If you open them *inside* ChimeraX (e.g. `open help:user/tutorials/binding-sites.html`, or **Help → Quick Start Guide**), the commands on the page become clickable links that run directly.

1) **Quick Start Guide** – a command-by-command tour: fetching a structure, styles and colours, selections, surfaces, biological assemblies, saving images; the second part deals with density maps. https://www.rbvi.ucsf.edu/chimerax/docs/quickstart/index.html
2) **Protein–Ligand Binding Sites** – the drug imatinib bound to Abl kinase (PDB 2HYY): atomic styles, hydrogen bonds and contacts, resistance mutations in the Sequence Viewer, surfaces coloured by lipophilicity and by electrostatic potential. https://www.rbvi.ucsf.edu/chimerax/docs/user/tutorials/binding-sites.html
3) **Coloring by Sequence Conservation** – opening a multiple sequence alignment, associating it with a structure (H-Ras, a GPCR, a porin) and colouring residues and surfaces by conservation, with a colour key. https://www.rbvi.ucsf.edu/chimerax/data/conservation-coloring/conservation-coloring.html

## EXERCISES

Repeat the VMD exercises with ChimeraX, save the images and compare them with those obtained with VMD.

1) ATP/$Mg^{2+}$ binding pose: `select (:ATP | ions) :<5; show sel atoms; view sel`. There are several copies of ATP: to focus on one, click it in the *Non-standard residues* table of the Log. Then identify the interactions with `hbonds :ATP reveal true` and `contacts :ATP reveal true` (the details are written in the Log). Which residues bind the phosphates? Where is the $Mg^{2+}$ ion?
2) DNA bending by ORC5 (chain E): `color /E:323-408 red`, then `contacts /E:323-408 restrict nucleic reveal true` to show which residues touch the DNA.
3) Using what you learned in tutorial 2, show the protein surface coloured by electrostatic potential (`coulombic protein`) and comment on the surface that faces the DNA.